Lección 3

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Plan de trabajo

Teoría de relatividad

– Cinemática Relativista y vectores 4-dimensionales
– Aplicaciones: la paradoja de gemelos, ...
– Dinámica Relativista
– Aplicaciones: ... efecto de Doppler relativista

▶ Nuestro Syllabus:

UNIDAD 1. Teoría de la Relatividad Especial
1.1. Transformaciones de Galileo
1.2. Experimento de Michelson y Morley
1.3. Postulados de Einstein
1.4. Transformaciones de Lorentz, Grupo de Lorentz
1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales
1.6. Cinemática Relativista
1.7. Dinámica Relativista
1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista

Recap: Special relativity

Albert Einstein, 1905:

The principle of relativity

The laws of physics apply (are equivalent) in all inertial reference frames

- Generalization of Galileo's principle from mechanics to all possible interactions

The universal speed of light

The speed of light in vacuum is the same in all reference frames.

- All special relativity is based on these principles
- The second postulate effectively replaces Galileo's transform with Lorentz transformations

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

– Time is not absolute, depends on frame
  ... simultaneity of events is not absolute
  ... time intervals are not constant
– Distance between moving objects depends on frame.

Recap: Lorentz group

Includes:

–Boosts in arbitrary direction:

$$\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{v^2}(\gamma-1) - \gamma t\right], \quad t' = \gamma\left(t - \frac{\boldsymbol{x}\cdot\boldsymbol{v}}{c^2}\right)$$

–Rotations:

$$\left(x'\right)^a = R^a{}_b x^b, \quad t' = t$$

Can rewrite in **compact matrix form**: $x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\boldsymbol{v})\, x^{\nu}$

$$\Lambda^{\mathrm{T}}\cdot\Lambda = \Lambda\cdot\Lambda^{\mathrm{T}} = 1 \qquad \Lambda \in \mathrm{SO}(3,1) \sim \mathrm{SO}(3)_{\mathrm{L}} \otimes \mathrm{SO}(3)_R$$

$$\Lambda^{\alpha}{}_{\mu}\Lambda_{\alpha}{}^{\nu} = \delta^{\nu}_{\mu}, \qquad \eta_{\alpha\beta}\Lambda^{\alpha}{}_{\mu}\Lambda_{\nu}{}^{\beta} = \eta_{\mu\nu}, \qquad \Lambda_{\mu}{}^{\alpha}(\boldsymbol{v}) = \left[\Lambda^{-1}(\boldsymbol{v})\right]^{\alpha}{}_{\mu} \equiv \Lambda^{\alpha}{}_{\mu}(-\boldsymbol{v})$$

- In general matrices $\Lambda^{\mu}{}_{\nu}$ might include a superposition of Lorentz boost & rotation
  - –Pure rotations form a rotations subgroub
  - –Pure boosts do NOT form a (sub)group
  - –NOT simple: is (locally) isomorphic to direct product $\sim \mathrm{SO}(3)_{\mathrm{L}} \otimes \mathrm{SO}(3)_{\mathrm{R}}$
  - –Dimension: 6, rank: 2.
  - –Elements of $\mathrm{SO}(3)_{\mathrm{L,R}}$: boosts in direction $\boldsymbol{v}$ <u>**and**</u> simultaneous (clockwise/counterclockwise) rotation around $\boldsymbol{v}$
- Group is non-commutative (non-abelian), $\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2 \cdot \Lambda_1$

Lorentz boosts: some applications

When You analyze different problems, You should carefully pay attention which transformation should be used: direct or inverse:

$$\Rightarrow x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

$$\Rightarrow x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}, \qquad t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

Assume the clock moves with constant velocity v. How the intervals of time measured in rest frame and lab frame are related to each other? How to get this using Lorentz transformation?

Note that in the lab frame the clock moves, and in the clock's rest frame, the laboratory moves with constant velocity.

Lorentz boosts: some applications

Relativistic time dilation (rest frame vs lab frame):

$$\Delta t_{\rm lab} = \frac{\Delta t_{\rm r.f.}}{\sqrt{1 - v^2/c^2}} \equiv \gamma(v)\Delta t_{\rm r.f.}$$

-time is dilated. Particle physics: for relativistic (*high-energy*) unstable particles their lifetime in lab-frame increases

Example: Muon decay

Assume that a high energy muon was detected to travel a distance L =800 meters in the lab frame before disintegrating. The lifetime of muon in the rest frame is $\tau = 2 \times 10^{-6} s$. According to naive estimate, the speed of muon is

$$v = \frac{L}{\tau} = 4 \times 10^8 m/s > c \quad (!!!)$$

Explain the reason of the paradox and estimate the velocity of muon.

Lorentz boosts: some applications

Solution:

In the lab-frame the lifetime of the muon is boosted by factor $\gamma(v)$, so the previous formula should be modified as

$$v = \frac{L}{\tau\gamma(v)} = \frac{L}{\tau}\sqrt{1 - \frac{v^2}{c^2}},$$

so the true velocity is given by

$$v = \frac{L}{\tau\sqrt{1 + L^2/\left(\tau^2 c^2\right)}} \approx 2.4 \times 10^8 m/s \approx 0.8\, c$$

no contradiction with 2nd postulate of Einstein (!)

Control problem - Twin's paradox

(Paul Langevin, 1911)

Assume that the first twin starts a trip in the rocket, whereas the second twin remains in the Earth. The rocket accelerates up to velocity v close to c to achieve a remote star, then turns around and returns back to the Earth. Which of the twins will be older on return ? Analyze how does the picture looks like from the point of view of both twins.

Control problem - Twin's paradox

From the lab on Earth:
(inertial frame)

- The time in the reference frame of the moving twin decelerated by factor $\langle\gamma(v)\rangle$

Assume for simplicity that:

- The distance to the star is d
- Magnitude of velocity is always constant v
- Direction might changes *instantly*.
- From the point of view of the twin on the Earth, the time of flight is

$$T = \frac{2d}{v}$$

▷Since the twin was in a moving frame, his time was Lorentz-dilated for the observer on the Earth, and on arrival he should report that the trip took

$$\tilde{T} = T\sqrt{1-v^2} = T/\gamma.$$

From the rocket's point of view

- The Earth moves with relativistic velocity v
  - distance $d$measured in the Earth's frame is contracted from the point of view of the moving twin,

$$\tilde{d} = d\sqrt{1-v^2} = \frac{d}{\gamma}$$

  - Length of the trip from the point of view of moving observer:

$$\tilde{T} = \frac{2\tilde{d}}{v} = \frac{2d}{v}\sqrt{1-v^2}$$

⇒No contradiction !!!

- More accurate analysis requires to take into account acceleration/deceleration, and the fact that the rocket's reference frame is <u>not</u> inertial ... However final result is the same

Lorentz boosts: some applications

When You analyze different problems, You should carefully pay attention which transformation should be used: direct or inverse:

$$\Rightarrow x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

$$\Rightarrow x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}, \qquad t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

A sphere of radius R in its rest frame is accelerated up to constant relativistic velocity $\boldsymbol{v}$ in the lab frame. What is the shape of the sphere seen by the observer in the lab frame? Try to get this using Lorentz transformation.

Lorentz boosts: some applications

Transformation of space distances depends on orientation:

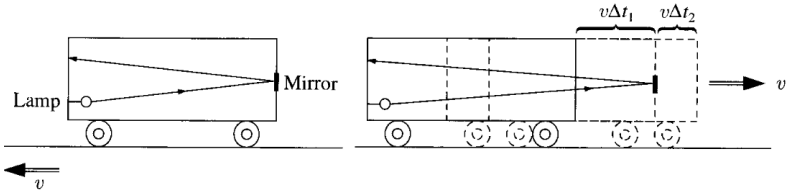
▷We should make *simultaneous* measurement (at the same time) at both ends
▷For transverse coordinates:

$$\Delta y' = \Delta y, \quad \Delta z' = \Delta z$$

▷For logitudinal coordinates:

$$\Delta L_{\rm lab} = \Delta L_{\rm r.f.} \sqrt{1 - v^2/c^2}$$

-Lorentz contraction of longitudinal distances



Assume that we define the length of the train measuring the time required for light to travel forth and back,

$$L = c\,(t_f - t_i)\,/2.$$

Evaluate the length L in different frames and demonstrate that the expressions (A, B) give correct transformation

Solution

▷ **Rest frame:**

– speed of light is c, walls at rest

– time for the ray to reach the mirror and return is

$$t_{\mathrm{r.f.}} = \frac{2L_{\mathrm{r.f.}}}{c} \Rightarrow L_{\mathrm{r.f.}} = \frac{ct_{\mathrm{r.f.}}}{2}$$

▷ **Lab frame:** the velocity of light is c, but:

– wall (mirror) moves away ("escapes") when the ray moves towards mirror, so time to reach the mirror is

$$t_1 = \frac{L_{\mathrm{lab}}}{c - v}$$

– wall (with lamp) moves towards the ray when the latter is reflected, so time to reach the wall is

$$t_2 = \frac{L_{\mathrm{lab}}}{c + v}$$

– Total time in lab frame:

$$t_{\mathrm{lab}} = t_1 + t_2 = \frac{2c\, L_{\mathrm{lab}}}{c^2 - v^2} \qquad (1)$$

– Rest frame time interval $t_{\mathrm{r.f.}}$ is dilated in lab frame,

$$t_{\mathrm{lab}} = \frac{t_{\mathrm{r.f.}}}{\sqrt{1 - v^2/c^2}} = \frac{2L_{\mathrm{r.f.}}}{c\sqrt{1 - v^2/c^2}} \qquad (2)$$

– From comparison of (1,2) conclude $L_{\mathrm{lab}} = L_{\mathrm{r.f.}}\sqrt{1 - v^2/c^2}$ as expected

Einstein's velocity addition

$$x = \frac{x' + Vt'}{\sqrt{1 - V^2/c^2}}, \qquad t = \frac{t' + \frac{Vx'}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

$$x' = \frac{x - Vt}{\sqrt{1 - V^2/c^2}}, \qquad t' = \frac{t - \frac{Vx}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

Assume that the particle moves with velocity $\boldsymbol{v}'$ in some reference frame R', and the reference frame moves in direction $\hat{\boldsymbol{x}}$ of the lab frame with constant velocity V. Find the velocity of particle in the lab frame. Take into account that the velocity $\boldsymbol{v}'$ might have "transverse" components (not collinear to $\hat{\boldsymbol{x}}$).

Einstein's velocity addition

We consider that time is dilated,

$$v_x = \frac{dx}{dt} = \frac{dx' + V\,dt'}{dt' + \frac{V\,dx'}{c^2}} = \frac{v_x' + V}{1 + \frac{v_x' V}{c^2}}$$

$$v_y = \frac{dy}{dt} = \frac{dy'\sqrt{1 - V^2/c^2}}{dt' + \frac{V\,dx'}{c^2}} = \frac{v_y'\sqrt{1 - V^2/c^2}}{1 + \frac{v_x' V}{c^2}}$$

▷For $c \to \infty$

$$v_x \approx v_x' + V, \quad v_y \approx v_y'$$

in agreement with Galileo's velocity addition rule

▷For $v_x' = c$, $v_y' = 0$ we get $v_x = c$, $v_y = 0$ in agreement with 2nd postulate

Einstein's velocity addition

Let's check what happens when light propagates in direction which forms angle $\alpha \neq 0$ with $\hat{\boldsymbol{x}}$:

▷For $v_x' = c\,\cos\alpha$, $v_y' = c\,\sin\alpha$ get

$$v_x = \frac{c\cos\alpha + V}{1 + \frac{V}{c}\cos\alpha}, \quad v_y = \frac{c\sin\alpha\sqrt{1 - V^2/c^2}}{1 + \frac{V}{c}\cos\alpha}$$

$$v_x^2 + v_y^2 = \frac{(c\cos\alpha + V)^2 + c^2\sin^2\alpha\left(1 - \frac{V^2}{c^2}\right)}{\left(1 + \frac{V}{c}\cos\alpha\right)^2} =$$

$$= \frac{\left(c^2 + V^2(1 - \sin^2\alpha) + 2Vc\cos\alpha\right)}{\left(1 + \frac{V}{c}\cos\alpha\right)^2} = c^2$$

–Speed of light in vacuum remains the same (in agreement with 2nd postulate), only changes direction of propagation:

–– Direction of propagation in lab frame <u>changes</u>:

$$\alpha_{\rm lab} = \arctan\left(\frac{v_y}{v_x}\right) = \arccos\left(\frac{v_x}{c}\right) \neq \alpha$$

Control problem 2

- Assume the particle moves with velocity $\boldsymbol{v}'$ in some reference frame R', and the reference frame moves in the lab frame with constant velocity $\boldsymbol{V}$ which does not coincide with $\hat{\boldsymbol{x}}$ What will be the velocity of the particle in the lab frame ?

Control problem 2

$$\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{V}\left[\frac{\boldsymbol{x}' \cdot \boldsymbol{V}}{V^2}(\gamma - 1) + \gamma t'\right], \quad t = \gamma\left(t' + \frac{\boldsymbol{x}' \cdot \boldsymbol{V}}{c^2}\right), \quad \gamma = \frac{1}{\sqrt{1 - V^2/c^2}}$$

$$\boldsymbol{v} = \frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{v}' + \boldsymbol{V}\left[\frac{\boldsymbol{v}' \cdot \boldsymbol{V}}{V^2}(\gamma - 1) + \gamma\right]}{\gamma\left(1 + \frac{\boldsymbol{v}' \cdot \boldsymbol{V}}{c^2}\right)}$$

Covariance of equations

The principle of relativity

The laws of physics apply (are equivalent) in all inertial reference frames

- An equation is said to be covariant if has the same form in any inertial reference frame. To guarantee this, we should use <u>only</u> covariant quantities
- Principle of covariance: All physical laws should be formulated using only those physical quantities the measurements of which the observers in different frames of reference could unambiguously correlate.

... Sounds like nice abstract words, but in reality You have already worked with "covariance" in 3 dimensions:

If we want to have equation which is isotropically invariant (keeps its form under rotations), should use vectors ($\boldsymbol{a}, \boldsymbol{b}, \ldots$) and operations which produce scalars, vectors ... ($\boldsymbol{a} \times [\boldsymbol{b} \times \boldsymbol{c}]$) but not their components explicitly, e.g.

$$\frac{d\boldsymbol{p}}{dt} = m\boldsymbol{g} + q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right) \text{ ...but not } \cancel{a_1 + b_2 + a_3 b_2}$$

Now we should do the same: operate with objects which transform "correctly" (covariantly) under Lorentz group transformations

Lorentz covariance of equations

The principle of relativity

The laws of physics apply (are equivalent) in all inertial reference frames

$$\Downarrow$$

- An equation is said to be Lorentz covariant if has the same form in any inertial reference frame. To guarantee this, we should use only Lorentz covariant quantities

$$\Downarrow$$

- =we should NOT mix covariant 4-vectors and 3-vectors (and avoid certain 3D constructions)

Our goal: learn how to write equations *covariantly*, using only objects which transform as tensors of some rank .

3D vs 4D covariant notations

Caution:

- Levi-Civita ε_{abc} is NOT Lorentz-covariant. Should replace it with 4D $\varepsilon_{\mu\nu\alpha\beta}$ or omit altogether/rewrite in terms of covariant structures.
  - ▷ Such definition is Lorentz-invariant

$$\frac{\partial x^\alpha}{\partial x'^{\mu'}}\frac{\partial x^\beta}{\partial x'^{\nu'}}\frac{\partial x^\gamma}{\partial x'^{\rho'}}\frac{\partial x^\delta}{\partial x'^{\lambda'}}\epsilon_{\alpha\beta\gamma\delta} = \epsilon_{\mu'\nu'\rho'\lambda'}\left|\frac{\partial x^\delta}{\partial x'^{\lambda'}}\right|$$

  and Jacobian $\left|\partial x^\delta/\partial x'^{\lambda'}\right| = 1$ as (recall proof of phase volume invariance)

– Instead of rotations around axis ϕ speak about rotation in plane

– Have *element of a surface (2-form, bivector)*

$$dA^{\mu\nu} = dx^\mu dy^\nu - dy^\mu dx^\nu$$

  formed by vectors dx^μ and dy^ν instead of "$d\boldsymbol{A} = \boldsymbol{n}\,|dA|$ *where* $\boldsymbol{n}$ *is normal to* $d\boldsymbol{x}$, $d\boldsymbol{y}$"

– Bivector ("antisym. tensor of rank 2")

$$A^\mu B^\nu - A^\nu B^\mu$$

  instead of *vector* $\boldsymbol{A}\times\boldsymbol{B}$ (recall that $[\boldsymbol{A}\times\boldsymbol{B}]_3 = A_1B_2 - A_2B_1$, ...)

Comparison: $3D \leftrightarrow 4D$ covariant objects

- Vector algebra operations in 4D:

	3D	**4D**
vector	$\vec{\boldsymbol{a}}$	$a^\mu,\ a_\mu = \eta_{\mu\nu} a^\nu$
scalar product	$\boldsymbol{a} \cdot \boldsymbol{b}$	$\eta_{\mu\nu} a^\mu b^\nu \equiv a_\mu b^\mu$
vector product	$(\boldsymbol{a} \times \boldsymbol{b})_i = \frac{\varepsilon_{ijk}}{2} (a_j b_k - a_k b_j)$	$a_\mu b_\nu - a_\nu b_\mu$
gradient	$\vec{\boldsymbol{\nabla}} f$	$\partial_\mu f$
div	$\nabla \cdot \vec{\boldsymbol{a}} \equiv \partial_k a^k$	$\partial_\mu f^\mu \equiv \eta^{\mu\nu} \partial_\mu f_\nu$
rot	$\left(\vec{\nabla} \times \vec{\boldsymbol{a}}\right)_i = \varepsilon_{ijk} \partial_j a_k$ $= \frac{1}{2} \varepsilon_{ijk} (\partial_j a_k - \partial_k a_j)$	$\partial_\mu a_\nu - \partial_\nu a_\mu$
Laplacian	$\Delta \equiv \nabla^2 \equiv \partial_k^2$	$\Box \equiv \partial_\mu \partial^\mu = c^{-2} \partial_t^2 - \Delta$

(see Goldstein, chapter 7 from our bibliography for more details)

Control question

- Equations of motion (EOM) in noncovariant (3D) notations.

$$\frac{d\boldsymbol{p}}{dt} = \boldsymbol{F} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right) \qquad (1)$$

–

Assume that a pointlike particle moves in external electromagnetic field $(\boldsymbol{E}, \boldsymbol{B})$. Please try to recall (if You remember) how the Lorentz force should be written in covariant notations?

Control question (reply)

- The covariant form is (FIS-221)

$$m\,\frac{du^\mu}{d\tau} \equiv \frac{dp^\mu}{d\tau} = q\,F^{\mu\nu}\,u_\nu \qquad (1)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$A^\mu = (\phi/c,\, \boldsymbol{A})\,, \quad A_\mu = \eta_{\mu\nu}A^\nu = (\phi/c,\, -\boldsymbol{A})\,.$$

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

– But we need to revisit how different objects (velocities, time intervals) transform under Lorentz transformations in order to understand why we get such structure

Covariance vs. Einstein's velocity addition

$$v_x = \frac{dx}{dt} = \frac{dx' + V\,dt'}{dt' + \frac{V\,dx'}{c^2}} = \frac{v'_x + V}{1 + \frac{v'_x V}{c^2}}$$

$$v_y = \frac{dy}{dt} = \frac{dy'\sqrt{1 - V^2/c^2}}{dt' + \frac{V\,dx'}{c^2}} = \frac{v'_y\sqrt{1 - V^2/c^2}}{1 + \frac{v'_x V}{c^2}}$$

- The velocity transformation looks complicated (nonlinear expressions).
  -Difficult to work with, NOT tensors.
- Most of the complexity (in denominators) due to time interval transformation $dt \to dt'$, dx'
- Assume that now we redefine all velocities as

$$u^\mu = dx^\mu / d\tau, \qquad (1)$$

where $d\tau$ is the proper time interval (in the rest frame of the partcle)

– Interval $d\tau$ is:
  -Lorentz scalar: time interval in rest frame of a particle
  -related to lab frame interval dt as $d\tau = dt\sqrt{1 - v^2/c^2}$

– All complexity related to denominator is gone. The object (1) will transform as covariant 4-vector (same as coordinates x^μ)

Proper 4-velocity

4-velocity (=Proper velocity)

$$u^\mu \equiv \frac{dx^\mu}{d\tau} = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{\vec{v}}{\sqrt{1 - v^2/c^2}} \right)$$

- The norm of the 4-velocity vector fulfils

$$u_\mu u^\mu \equiv \eta_{\mu\nu} u^\mu u^\nu = c^2$$

- The 4-velocity transforms as components of 4-vector (same as for coordinates) :

$$u^\mu \to \Lambda^\mu_\nu u^\nu$$

Special case: boost in $\hat{x}$-direction:

$$\begin{aligned}
\left(u^1\right)' &= \gamma \left(u^1 - v u^0\right) \\
\left(u^0\right)' &= \gamma \left(u^0 - \frac{v u^1}{c^2}\right) \\
\left(u^\mu\right)' &= \Lambda^\mu_\nu u^\nu
\end{aligned}$$

Example

Assume that a particle moves along $\hat{\boldsymbol{x}}$ direction, and its trajectory is given by

$$x(t) = \sqrt{a^2 + c^2 t^2}, \quad y = z = 0$$

(hyperbolic motion in $x - t$ plane). We'll evaluate the instant velocity $\vec{\boldsymbol{v}}$ and the 4-velocity u^μ for this case

$$v_x = \frac{dx}{dt} = \frac{c^2 t}{\sqrt{a^2 + c^2 t^2}},$$

$$v_y = v_z = 0$$

$$\sqrt{1 - \frac{v^2}{c^2}} \equiv \frac{a}{\sqrt{a^2 + c^2 t^2}}$$

$$u^\mu = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right) = \frac{c}{a} \left(\sqrt{a^2 + c^2 t^2}, ct \right)$$

Proper 4-momentum

Covariant 4-momentum

$$p^\mu \equiv m\, u^\mu = \left(\frac{mc}{\sqrt{1 - v^2/c^2}}, \frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right),$$

in $v \ll c$ limit

$$\frac{mc}{\sqrt{1 - v^2/c^2}} \approx mc \left(1 + \frac{v^2}{2\, c^2} \right) \quad (1)$$

$$\approx \frac{1}{c} \left(mc^2 + \underbrace{\frac{mv^2}{2}}_{\text{kinetic energy}} \right)$$

$$\frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \approx \underbrace{m\vec{\boldsymbol{v}}}_{\text{momentum}} \quad (2)$$

$\Rightarrow$Identify components of p^μ with energy-momentum,

$$p^\mu = \left(\frac{E}{c}, \vec{p} \right), \quad E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

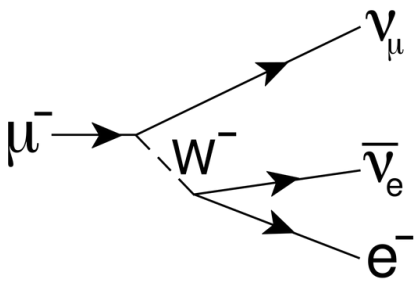
Proper 4-momentum

$$p^\mu = \left(\frac{E}{c}, \vec{p}\right) \approx \frac{1}{c}\left(mc^2 + \underbrace{\frac{mv^2}{2}}_{\text{kinetic energy}}, \underbrace{m\vec{v}}_{\text{momentum}}\right)$$

- The particle at rest has (huge!) energy

$$E_{\text{rest}} = mc^2$$

- Was strange at the beginning of 20th century, now fully confirmed:
  - Decay of a muon into fragments, $m_\mu \gg \sum m_i$. Rest frame energy converts into kinetic energy of fragments



- Massive particle ($m \neq 0$) cannot be accelerated up to $v \geq c$, energy $E \to \infty$
- The ratio $|p|/E = v/c^2$, so is we assume that some particle might move with $v \approx c$ (*ultrarelativistic limit*), then

$$E = |\vec{p}|\, c$$

Transformation of proper 4-momentum

4-momentum (definition)

$$p^\mu \equiv m\,u^\mu = \left(\frac{mc}{\sqrt{1-v^2/c^2}}, \frac{m\vec{\boldsymbol{v}}}{\sqrt{1-v^2/c^2}} \right),$$

► By construction the components of the 4-vector p^μ transform as coordinates x^μ and 4-velocity u^μ, viz

$$p'^\mu = \Lambda^\mu{}_\nu p^\nu$$

–boost in x-direction (replace $ct \to E/c$, $\vec{\boldsymbol{x}} \to \vec{\boldsymbol{p}}$):

$$E' = \frac{E - V\,p_x}{\sqrt{1-V^2/c^2}}, \quad p'_x = \frac{p_x - V\,E/c^2}{\sqrt{1-V^2/c^2}}, \quad p'_y = p_y, \quad p'_z = p_z$$

–boost in arbitrary direction:

$$\boldsymbol{p}' = \boldsymbol{p} + \boldsymbol{v}\left[\frac{\boldsymbol{p}\cdot\boldsymbol{v}}{v^2}(\gamma-1) - \gamma\frac{E}{c^2}\right], \quad E' = \gamma\,(E - \boldsymbol{p}\cdot\boldsymbol{v})$$

Transformation law for the 3-force

Lorentz boosts (in x-direction):

$$x = \frac{x' + Vt'}{\sqrt{1 - V^2/c^2}}, \qquad t = \frac{t' + \frac{Vx'}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

$$x' = \frac{x - Vt}{\sqrt{1 - V^2/c^2}}, \qquad t' = \frac{t - \frac{Vx}{c^2}}{\sqrt{1 - V^2/c^2}}, \quad y' = y, \quad z' = z$$

Transformation for the force is quite sophisticated:

$$F'_y = \frac{dp'_y}{dt'} = \frac{F_y}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

$$F'_z = \frac{dp'_z}{dt'} = \frac{F_z}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

$$F'_x = \frac{dp'_x}{dt'} = \frac{F_x - (v_x/c)\,(dE/dt)}{1 - v_x\, v_{\mathrm{body}}/c^2} = \frac{F_x - (v_x/c)\,(F \cdot v)}{1 - v_x\, v_{\mathrm{body}}/c^2}$$

4-force

Assume that we defined a 4-vector of force

$$g^\mu = dp^\mu / d\tau$$

How it transforms under Lorentz boosts ?
Evaluate its components in terms of the 3D force $\boldsymbol{f} = d\vec{\boldsymbol{p}}/dt$.
Demonstrate that $u \cdot g \equiv u_\mu g^\mu \equiv \eta_{\mu\nu} u^\mu g^\nu = 0$.

4-force (II)

Components in terms of 3-force:

$$g^\mu = \left(\frac{\boldsymbol{f} \cdot \boldsymbol{v}}{c\sqrt{1 - v^2/c^2}}, \frac{\boldsymbol{f}}{\sqrt{1 - v^2/c^2}} \right)$$

– Difference from $d\boldsymbol{p}/dt$ by factor $dt/d\tau = 1/\sqrt{1 - v^2/c^2}$
– Recall that $\boldsymbol{f} \cdot \boldsymbol{v}$ is the work done per unit of time, so we understand why it is 0th component.
– Transformation under Lorentz boost like any 4-vector:

$$\boldsymbol{g}' = \boldsymbol{g} + \boldsymbol{v} \left[\frac{\boldsymbol{g} \cdot \boldsymbol{v}}{v^2}(\gamma - 1) - \gamma \frac{g^0}{c} \right], \quad g_0' = \gamma \left(E - \boldsymbol{p} \cdot \boldsymbol{v} \right)$$

$$\eta_{\mu\nu} u^\mu u^\nu = 1,$$

Apply $d/d\tau$ to both parts, recall that $p^\mu = m\, u^\mu$

$$\Rightarrow u \cdot g = 0$$

4-force

►In what follows we may write equations of motion (EOM) in lab frame, using non-covariant derivative d/dt and force

$$\frac{d\boldsymbol{p}}{dt} = \boldsymbol{F} \quad (1)$$

or use covariant Minkowski 4-force $dp^\mu/d\tau$,

$$\frac{dp^\mu}{d\tau} = f^\mu_{\text{external}} \quad (2)$$

But we should always cross-check that both parts of EOM transform *covariantly* (in the same way).

►Example: Lorentz force:

$$\vec{\boldsymbol{F}} = q\left(\vec{\boldsymbol{E}} + \vec{\boldsymbol{v}} \times \vec{\boldsymbol{B}}\right)$$

-is 3-vector (should use (1)), but we may rewrite it as

$$f^\mu = q\, F^{\mu\nu} u_\nu$$

where $F_{\mu\nu}$ is the tensor of electromagetic field and u_ν is the 4-vector

Caution with relativistic system

Relativistic system is not just modification of energy/momentum

►some concepts/qualitative features are *different* from $v \ll c$ limit

We no longer have long-distance interaction

– no $U(\boldsymbol{r}_1 - \boldsymbol{r}_2)$-type terms are allowed in the action

–For closed system should consider interaction *in the same point* x^μ, field+particle.

– MUST include terms both for particles and fields to keep covariance

►Field (e.g. electromagnetic) carries away part of energy-momentum, will affect balance of energy-momentum

Can't introduce concept of the center-of-mass

$$\vec{R}_{\mathrm{CM}}(t) = \frac{\sum_{i=1}^{N} m_i \, \vec{x}_i(t)}{\sum_{i=1}^{N} m_i}$$

and "separate" its motion

►Collective momentum $\boldsymbol{P} = \sum \boldsymbol{p}_i$ has nothing to do with $d\vec{R}_{\mathrm{CM}}(t)/dt$.

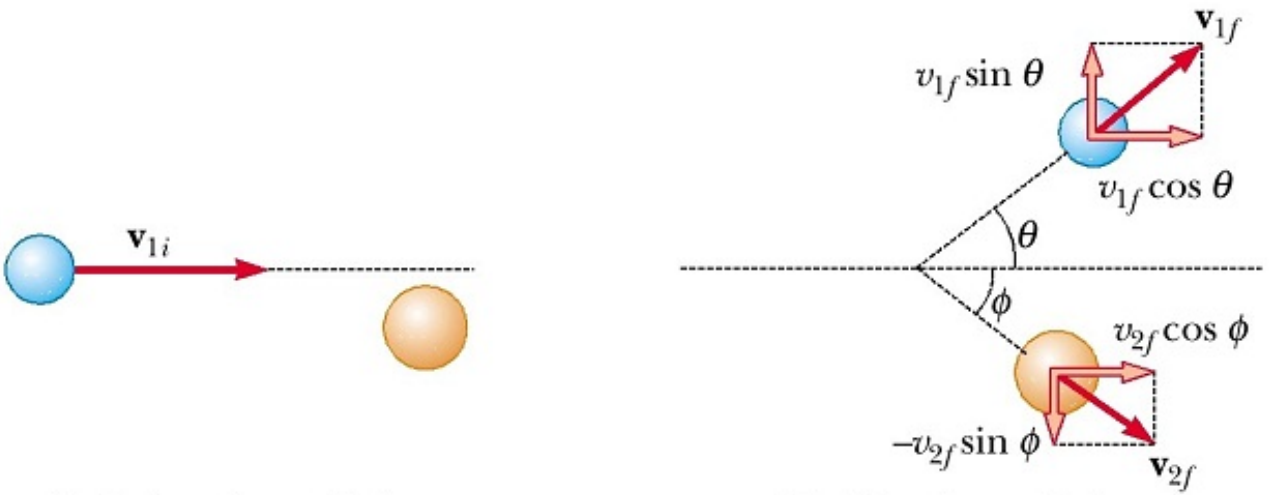
►We can choose "C-frame" in which $\boldsymbol{P} = 0$, but it is not CM frame!

Energy-momentum conservation - Control question

- So far we introduced relativistic 4-momenta as extensions of nonrelativistic variables which transform "correctly" (covariantly)
- *Assume we have a system of particles, and try to apply energy-momentum conservation, as expected from the Noether's theorem. Which quantities are conserved,*

$$\boldsymbol{p}_{\text{nonrel}} = \sum m_a \boldsymbol{v}_a \qquad \text{or} \qquad \boldsymbol{p}_{\text{rel}} = \sum \frac{m_a \boldsymbol{v}_a}{\sqrt{1 - v_a^2/c^2}}?$$

$$E_{\text{nonrel}} = \sum \frac{m_a v_a^2}{2} \qquad \text{or} \qquad E_{\text{rel}} = \sum \frac{m_a c^2}{\sqrt{1 - v_a^2/c^2}}?$$



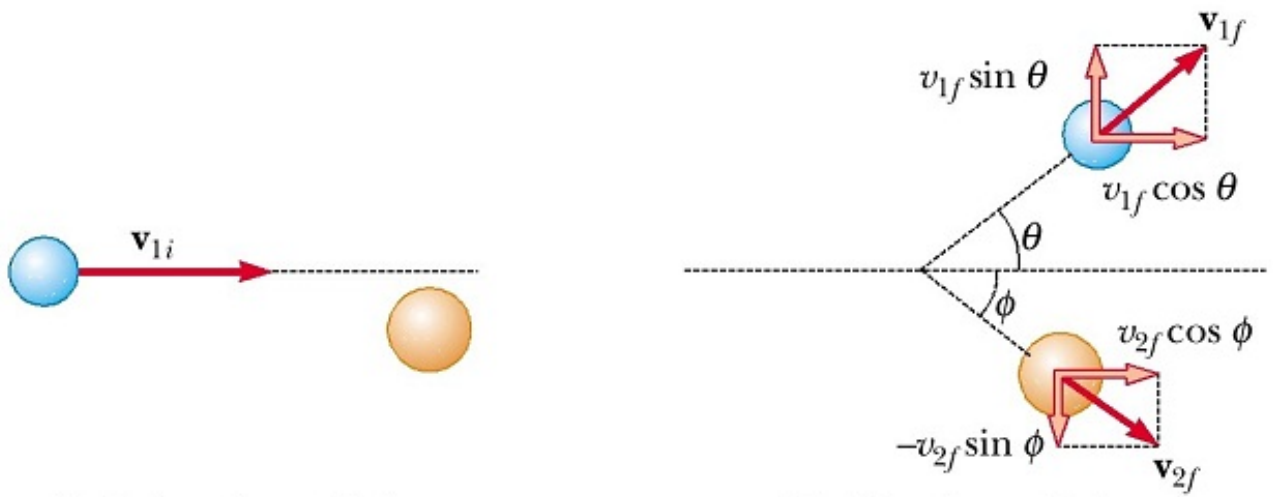
(a) Before the collision (b) After the collision

Energy-momentum conservation

- *If we have a system of particles, then we have energy-momentum conservation, as expected from Noether theorem. However, we should use* <u>*relativistic*</u> *expressions for them*
- *We may write energy-momentum conservation in relativistic notations, using 4-vectors as*

$$\sum_n p^\mu_{(n)} = \text{const}$$

- *In the next lecture we'll see in detail how this works, consider elastic collision* $2 \to 2$*, and learn how to work efficiently with 4-momenta.*



(a) Before the collision (b) After the collision

Decay of a particle

Assume that a particle of mass M decays into two fragments of masses m_1 and m_2 (obviously, $M > m_1 + m_2$).

1) Find the energies of the partciles in the rest frame of the initial particle ("C-frame").
2) Assume that the initial particle moves in the laboratory with velocity V, and after the decay, the fragments follow straight lines which form angles θ_1, θ_2 with trajectory of the initial particle. Evaluate the relation between angles θ_1, θ_2 and the energies of the particles

Decay of a particle (solution)

Energy:momentum conservation:

$$M\,c^2 = E_1 + E_2, \ (1) \qquad 0 = \boldsymbol{p}_1 + \boldsymbol{p}_2 \ (2) \qquad \boldsymbol{p}_2 = -\boldsymbol{p}_2 \ (2')$$

$$E_{1,2} = c\,\sqrt{p_{1,2}^2 + m_{1,2}^2 c^2}$$

$$\Rightarrow \frac{E_1^2}{c^2} - m_1^2 c^2 = \frac{E_2^2}{c^2} - m_2^2 c^2 \quad (3)$$

$$\Rightarrow \frac{E_1^2 - E_2^2}{c^2} = \frac{(E_1 - E_2)\,(E_1 + E_2)}{c^2} = \left(m_1^2 - m_2^2\right) c^2 \quad (3')$$

From (1) and (3')

$$\Rightarrow E_1 = \frac{M^2 + m_1^2 - m_2^2}{2\,M}, \qquad E_2 = \frac{M^2 - m_1^2 + m_2^2}{2\,M}$$

Decay of a particle (solution)

We know that in C-frame and lab-frame the energies are bound by

$$E_1^{(\mathrm{C})} = \frac{E_1^{(\mathrm{lab})} - V\, p^{(\mathrm{lab})} \cos\theta}{\sqrt{1 - V^2/c^2}}$$

where $E_1^{(\mathrm{C})}$ was found earlier, depends on $m_{1,2}$ only.

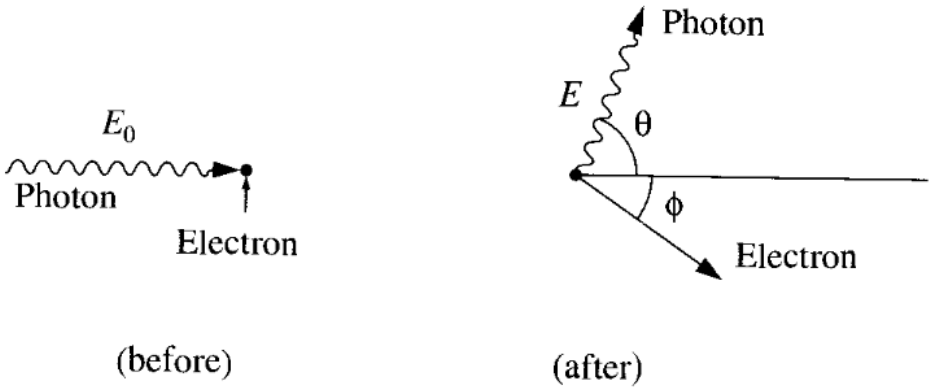
$$E_1^{(\mathrm{lab})} = c\sqrt{\left(p^{(\mathrm{lab})}\right)^2 + m_1^2 c^2},$$

$$\Rightarrow \cos\theta = \frac{E_1^{(\mathrm{lab})} - E_1^{(\mathrm{C})}\sqrt{1 - V^2/c^2}}{V\, p^{(\mathrm{lab})}} = \frac{E_1^{(\mathrm{lab})} - E_1^{(\mathrm{C})}\sqrt{1 - V^2/c^2}}{V\sqrt{\left(E_1^{(\mathrm{lab})}/c\right)^2 - m_1^2 c^2}}$$

Can solve algebraically for $E_1^{(\mathrm{lab})}$ for a given θ_1. Similarly for the second particle

Compton scattering

Photon with energy E_0 scatters on electron, initially at rest. Find the relation between the energy of the scattered photon and the scattering angle θ



Caution:

- In *classical* electrodynamics we work with EM fields $\boldsymbol{E}$, $\boldsymbol{B}$
- Concept of photon appears in *quantum* electrodynamics. Now consider it simply as *ultrarelativistic particle* with energy-mass relation $E = |\boldsymbol{p}|\, c$ (moving with speed of light c).
- Relation to the wavelength of EM wave:

$$\lambda := \frac{h\, c}{E}$$

Compton scattering

►We'll work in lab-frame, use energy-momentum conservation

–Use notations p_e, p_p for momenta of scattered electron and photon

After scattering:

$$p_0 = p_e \cos\phi + p_p \cos\theta \quad (1)$$

$$p_e \sin\phi = p_p \sin\theta \quad (2)$$

$$p_0 c + mc^2 = p_p c + c\sqrt{p_e^2 + m^2c^2} \quad (3)$$

-system of 3 equations with unknown $p_p = |E_e|/c$, ϕ, p_e, use algebraic methods to solve:

$$p_0 - p_p \cos\theta = p_e \cos\phi \quad (1')$$

$$\Rightarrow (p_0 - p_p - mc)^2 = p_e^2 + m^2c^2 \quad (3')$$

$(1')^2 + (2)^2$:

$$(p_0 - p_p \cos\theta)^2 + p_p^2 \sin^2\theta = p_e^2$$

$$p_0^2 + p_p^2 - 2p_0 p_p \cos\theta = p_e^2 \quad (4)$$

$(4) - (3')$:

$$2p_0 p_p (1 - \cos\theta) + 2mc(p_0 - p_p) = 2m^2c^2$$

Take into account $E_p = p_p c$, $E_0 = p_0 c$ and solve equation:

$$E_p = p_p c = \frac{1}{(1 - \cos\theta)/mc^2 + 1/E_0}$$

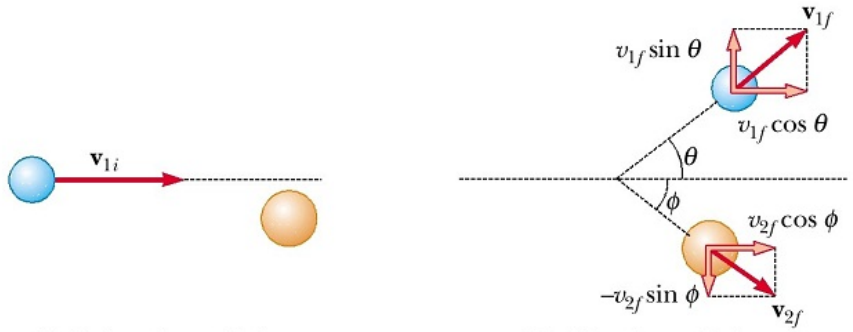
$$\lambda = \lambda_0 + \underbrace{\frac{h}{mc}}_{\lambda_e}(1 - \cos\theta)$$

λ_e-wavelength of electron

Nonrelativistic elastic scattering

(***Classical mechanics***)

►Consider $2 \to 2$ elastic scattering



(a) Before the collision (b) After the collision

- Energy-momentum conservation: 4 independent constraints
  $\Rightarrow$if we know ($\boldsymbol{p}_{1\,I}$, $\boldsymbol{p}_{2\,I}$), and only one of the angles (θ_1, θ_2) or energies *after* scattering (E_1, E_2), can reconstruct the remaining variables

Center of mass frame

1) The scattering always happens in a plane
2) The absolute values of momenta are conserved, $|\boldsymbol{p}_{1\,I}| = |\boldsymbol{p}_{1\,F}| = |\boldsymbol{p}_{2\,I}| = |\boldsymbol{p}_{2\,F}|$
3) The scattering angles are equal, $\theta_1 = \theta_2$
4) The relative motion:

$$\boldsymbol{r}_{12} = \boldsymbol{r}_1 - \boldsymbol{r}_2$$

with reduced mass
$\mu = m_1 m_2/(m_1 + m_2)$

Relativistic elastic scattering

In **relativistic case** there are also relations dictated by energy-momentum conservation. We'll derive them, and will use only covariant notations (in order to get covariant results):

In relativistic case angles, energies transform using expression which are significantly more complicated than in nonrelativistic case

It might be easier/faster to formulate possible kinematic constraints using invariant <u>*scalar products of 4-vectors*</u> $p_a \cdot p_b \equiv \eta_{\mu\nu} p_a^\mu p_b^\nu$, *and later just write out these 4-momenta in terms of the angles/energies in the frame which is of interest for us.*

Relativistic elastic scattering

►4-momentum conservation:

$$p^{\mu}_{1i} + p^{\mu}_{2i} = p^{\mu}_{1f} + p^{\mu}_{2f} \quad (1)$$

-4 equations

►"Onshellness conditions" $p^2_{1,2} = m^2_{1,2}$ $(p^2 \equiv E^2/c^2 - p^2)$

– ... should write for all 4 momenta $(p_{1i,\,2i,\,1f,\,2f})$

►Numerous relations:

$$(p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$

$$\Rightarrow \underbrace{p^2_{1i}}_{m^2_1} + \underbrace{p^2_{2i}}_{m^2_2} + 2p_{1i} \cdot p_{2i} = \underbrace{p^2_{1f}}_{m^2_1} + \underbrace{p^2_{2f}}_{m^2_2} + 2p_{1f} \cdot p_{2f}$$

$$\Rightarrow p_{1i} \cdot p_{2i} = p_{1f} \cdot p_{2f}$$

$$(p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$

$$\Rightarrow \underbrace{p^2_{1f}}_{m^2_1} + \underbrace{p^2_{1i}}_{m^2_1} - 2p_{1f} \cdot p_{1i} = \underbrace{p^2_{2f}}_{m^2_2} + \underbrace{p^2_{2i}}_{m^2_2} - 2p_{2f} \cdot p_{2i}$$

$$\Rightarrow m^2_1 - p_{1f} \cdot p_{1i} = m^2_2 - p_{2f} \cdot p_{2i}$$

Relativistic elastic scattering

- 4-momentum conservation:

$$p_{1i}^{\mu} + p_{2i}^{\mu} = p_{1f}^{\mu} + p_{2f}^{\mu} \quad (1)$$

- Consequences:

$$(p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$
$$\Rightarrow p_{1f} \cdot p_{2i} = p_{2f} \cdot p_{1i}$$

$$p_{2f}^2 = (p_{1i} + p_{2i} - p_{1f})^2$$
$$\Rightarrow \cancel{m_2^2} = 2\, m_1^2 + \cancel{m_2^2} + 2p_{1i} \cdot p_{2i}$$
$$- 2p_{1i} \cdot p_{1f} - 2p_{2i} \cdot p_{1f}$$

$$\Rightarrow p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f}$$

substitution ($i \leftrightarrow f$):

$$\Rightarrow p_{1f} \cdot p_{2f} + m_1^2 = p_{1i} \cdot p_{1f} + p_{1i} \cdot p_{2f}$$

substitution ($1 \leftrightarrow 2$):

$$\Rightarrow p_{1i} \cdot p_{2i} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f}$$

substitution ($1 \leftrightarrow 2$, $i \leftrightarrow f$):

$$\Rightarrow p_{1f} \cdot p_{2f} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f}$$

Relativistic elastic scattering

$$p_{1i} \cdot p_{2i} = p_{1f} \cdot p_{2f} \quad (2)$$

$$m_1^2 - p_{1f} \cdot p_{1i} = m_2^2 - p_{2f} \cdot p_{2i} \quad (3)$$

$$p_{1f} \cdot p_{2i} = p_{2f} \cdot p_{1i} \quad (4)$$

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

$$p_{1f} \cdot p_{2f} + m_1^2 = p_{1i} \cdot p_{1f} + p_{1i} \cdot p_{2f} \quad (6)$$

$$p_{1i} \cdot p_{2i} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f} \quad (7)$$

- Let's assume one of the particles is at rest, $p_{2i} = (m_2 c, 0)$:

$$p_{1i} = \left(\sqrt{p_1^2 + m^2 c^2}, \, p_1 \right), \quad |\boldsymbol{p}_1| = \sqrt{E_1^2/c^2 - m_1^2 c^2}$$

$$p_{1i} \cdot p_{2i} = E_{1i} \, m_2 c, \quad p_{2i} \cdot p_{1f} = m_2 c \, E_{1f}, \quad p_{1i} \cdot p_{1f} = E_{1i} E_{1f} - |\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}| \cos\theta_1$$

From (5)

$$\cos\theta_1 = \frac{E_{1f}/c \, (E_{1i}/c + m_2 c)}{|\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}|} - \frac{E_{1i} m_2 c + m_1^2 c^2}{|\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}|}$$

Mandelstam invariants - Elastic scattering

Previous results:

- Energy of initial & scattered particles fully fix kinematics (angles)
- Note that energies/angles depend essentialy on reference frames, not very convenient
- Now we'll show that Mandelstam invariants (variables)

$$s := (p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$
$$t := (p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$
$$u := (p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$

are sufficient to fix the kinematics of the process $2 \to 2$.

Lemma

Assume that we have some **inelastic** process

$$1 + 2 \to 3 + 4.$$

Masses of all particles 1,2,3,4 are different. Then Mandelstam variables s, t, u are bound by

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

Proof:
Momentum conervation:

$$p_1 + p_2 - p_3 - p_4 = 0 \quad (1)$$

$$s + t + u = (p_1 + p_2)^2 + (p_1 - p_3)^2 + (p_1 - p_4)^2 \quad (2)$$
$$= (p_3 + p_4)^2 + (p_2 - p_4)^2 + (p_2 - p_3)^2 \quad (3)$$

– Let's sum (2+3) and take into account that $p_i^2 = m_i^2$, as well as identities

$$3p_i^2 = 2m_i^2 + p_i \cdot p_i$$
$$2p_i \cdot p_j = p_i \cdot p_j + p_j \cdot p_i$$

Lemma

$$
\begin{aligned}
&2\left(s+t+u\right)= \\
&=2\left(m_1^2+m_2^2+m_3^2+m_4^2\right) \\
&+p_1\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&+p_2\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&-p_3\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&-p_4\cdot\left(p_1+p_2-p_3-p_4\right)= \\
&2\left(m_1^2+m_2^2+m_3^2+m_4^2\right)
\end{aligned}
$$

$\Rightarrow$Conclude that (A) is correct, Q.E.D.

Mandelstam variables - Elastic scattering

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

- (invariant) Mandelstam variables:

$$s := (p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$
$$t := (p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$
$$u := (p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$

Common practice: express all physical observables/cross-sections in terms of s, t, u instead of jumping from one frame to another

Elastic scattering:

$$s + t + u = 2\left(m_1^2 + m_2^2\right)$$

-only 2 of them are independent

Rest frame of one of the particles:

$$s = (E_{1i} + m_2)^2 - p_1^2 = m_1^2 + m_2^2 + 2\,E_{1i} m_2$$

s is related to energy of collision

$$t = 2\left(m_1^2 - p_{1i} \cdot p_{1f}\right)$$

-"absolute value" of momentum transfer from one particle to another; from (5)

$$p_{1i} \cdot p_{1f} = E_{1i} m_2 + m_1^2 - m_2 E_{1f}$$
$$= m_1^2 + m_2\left(E_{1i} - E_{1f}\right)$$

so

$$t = -2m_2\left(E_{1i} - E_{1f}\right) \leq 0$$

is related to energy change of one particle. From

$$\left(m_1^2 - \frac{t}{2}\right) = p_{1i} \cdot p_{1f} = E_{1i} E_{1f} -$$
$$- \left|\boldsymbol{p}_{1i}\right| \left|\boldsymbol{p}_{1f}\right| \cos\theta_1$$

$$\left|\boldsymbol{p}_1\right| = \sqrt{E_1^2/c^2 - m_1^2 c^2}$$

can find scattering angles θ_1 of elastic scattering

Mandelstam variables-Exercise

Assume that we have some inelastic process

$$1 + 2 \to 3 + 4.$$

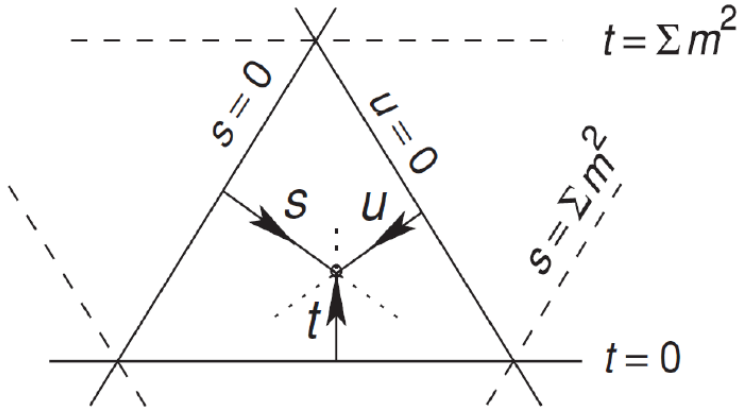
Masses of all particles 1,2,3,4 are different. Then Mandelstam variables s, t, u are bound by

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

▷Relation (A) implies that only 2 variables are independent

▷ Viviani's theorem: for any point inside equilateral triangle sum of distances equals height of triangle

⇒Graphical illustration of relation (A):



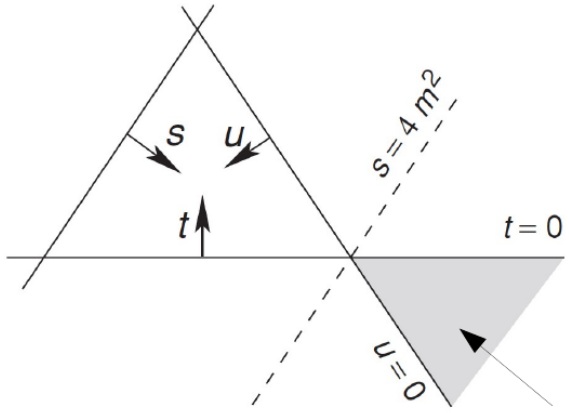
Assume for simplicity that:

▷ all masses are equal, $m_i \equiv m$

▷ work in CM frame of particles $1, 2$, so

$$p_1 = (E, \boldsymbol{p}),$$
$$p_2 = (E, -\boldsymbol{p}),$$
$$E = \sqrt{m^2 + \boldsymbol{p}^2}$$

$$s = (p_1 + p_2)^2 =$$
$$= 2m^2 + 2\left(E^2 + \boldsymbol{p}^2\right) =$$
$$= 4m^2 + 4\boldsymbol{p}^2 \geq 4m^2$$



physical region of scattering

Mandelstam variables-Exercises

Assume that we study process

$$1 + 2 \to 3 + 4.$$

Masses of all particles 1,2,3,4 are different. Prove that the Mandelstam variables s, t, u are bound by inequalities

$$s \geq (m_1 + m_2)^2 , \quad (A)$$
$$s \geq (m_3 + m_4)^2 , \quad (B)$$
$$t \leq (m_2 - m_4)^2 , \quad (C)$$
$$t \leq (m_1 - m_3)^2 , \quad (D)$$
$$u \leq (m_1 - m_4)^2 , \quad (E)$$
$$u \leq (m_2 - m_3)^2 . \quad (F)$$

Mandelstam variables-Exercises

Assume we study process

$$1 + 2 \to 3 + 4.$$

and we've chosen 3 independent momenta q_1, q_2, q_3 (the 4th might be found from energy-momentum conservation) and consrtuct vector

$$L_\alpha = \varepsilon_{\alpha\beta\mu\nu} q_1^\beta q_2^\mu q_3^\nu .$$

Show that in the rest frame of particle $1, 2$ or 3 the vector L_α cannot have 0-component, and for this reason

$$L_\alpha L^\alpha \equiv -\boldsymbol{L}^2 \leq 0 \quad (A)$$

Show that in arbitary frame

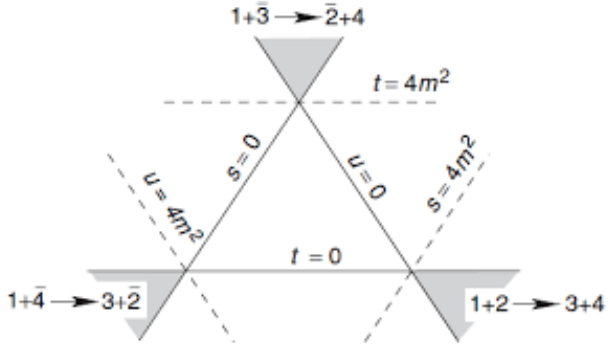
$$-L_\alpha L^\alpha = \det \begin{vmatrix} q_1^2 & q_1 \cdot q_2 & q_1 \cdot q_3 \\ q_1 \cdot q_2 & q_2^2 & q_2 \cdot q_3 \\ q_1 \cdot q_3 & q_2 \cdot q_3 & q_3^2 \end{vmatrix} = \det \begin{vmatrix} m_1^2 & \frac{s - m_1^2 - m_2^2}{2} & \frac{m_1^2 + m_3^2 - t}{2} \\ \frac{s - m_1^2 - m_2^2}{2} & m_2^2 & \frac{m_2^2 + m_3^2 - u}{2} \\ \frac{m_1^2 + m_3^2 - t}{2} & \frac{m_2^2 + m_3^2 - u}{2} & m_3^2 \end{vmatrix} \geq 0$$

-this gives nontrivial conditions for possible values of s, t, u (*Kibble*)

Mandelstam variables

Further reading:

- *Problems* in Landau, Vol. 2, Section 13
- Vol. 4 of Berestetskii, V.B., Lifshitz, E.M. and Pitaevskii, L.P. (1974) Relativistic Quantum Theory. Pergamon Press, Section 66-67
- If You will try to seek Mandelstam variables on your own, You may find something like:



– in Quantum Field Theory it is shown that the same cross-section (function) describes the process $1 + 2 \to 3 + 4$, as well as processes which might be obtained replacing particle with momentum p in initial state with its antiparticle with momentum $-p$ in final state (and vice versa). This leads to permutation of variables s, t, u and explains why we get into "unphysical" region (In the above-given plot $\bar{a}$ implies antiparticle of particle a).